
IAL Chemistry Unit 2

Rapid Revision

WCH12 — Energetics, Group Chemistry, Halogenoalkanes and Alcohols

Last-minute cram sheets — designed to be scanned, not read

Condensed from the full A* guide (18 past papers, Jan 2020 - Jun 2026)

80 marks · 1 h 30 min · $\approx$ 1.1 min per mark

Section A: 20 MCQ (always exactly 20) · Sections B + C: always 60

One 6-mark asterisked (*) QER question, every paper

TRIAGE — WHERE THE MARKS ARE

Appears in 16-18 of 18 papers — non-negotiable

Intermolecular forces · oxidation numbers · halogenoalkane substitution/elimination · Group 2 thermal stability · equilibrium and Le Chatelier · infrared spectra · chlorine chemistry

Appears in 10-15 of 18 — very likely, they rotate

Calorimetry · enthalpy definitions · bond enthalpies · Hess cycles · redox half-equations · disproportionation · flame tests · conc. H_2SO_4 with halides · silver nitrate tests · titrations · Maxwell-Boltzmann · mechanisms · alcohol oxidation · mass spectra · gas volumes · significant figures

THE FREE MARKS YOU MUST NOT LOSE

State symbols — "include state symbols" loses a whole mark if omitted; appears almost every session.

Sign + units + significant figures — these are **separately awarded** marking points. A right number with the wrong sign, no units, or 5 s.f. from 3 s.f. data still loses marks.

Never leave a calculation blank — **TE (transferred error)** runs through most multi-step calculations, so an early slip rarely costs all the marks. Show every line.

THE 6-MARK QER — HOW IT IS ACTUALLY MARKED

Two separate passes, added together:

Indicative content (3-4 marks) — a checklist of ~5-6 chemistry points. **Order and phrasing do not matter, only how many you hit.**

Structure and reasoning (2 marks) — are your points **linked** with *because*, *therefore*, *this means that*, or just a disconnected list?

5-6 points + fully linked = 4 + 2. 3-4 points + some linkage = 2 + 1. 1-2 points, no linkage = 1 + 0.

So: brain-dump 5-6 separate points first, **then** spend 30 seconds inserting connectives. Coverage beats elegance. **A wrong statement costs the reasoning marks** — leave out anything you are unsure of.

SHEET 1 — ENERGETICS

Enthalpy change	Definition — every one needs "1 mole", standard conditions (100 kPa, stated T) and standard states
Formation	when 1 mol of a compound is formed from its elements in their standard states
Combustion	when 1 mol of a substance is completely combusted in excess oxygen
Atomisation	when 1 mol of gaseous atoms is formed from the element in its standard state
Neutralisation	when 1 mol of water is formed from an acid and an alkali
Reaction	for the molar quantities as written in the equation — the only one not per mole of a substance
Solution	when 1 mol of solute dissolves completely to an infinitely dilute solution

Section A trap: spot the wrong equation — a fractional coefficient giving the wrong product state, an element **not in its standard state** (Cl(g) instead of $\frac{1}{2}\text{Cl}_2(\text{g})$), or **2 mol of product** instead of 1. Always check: exactly 1 mole of the defined species, all elements in their standard physical state.

HESS — GET THE DIRECTION RIGHT OR EVERY SIGN FLIPS

Given **formation** data: $\Delta_r H = \Sigma \Delta_f H(\text{products}) - \Sigma \Delta_f H(\text{reactants})$

Given **combustion** data: $\Delta_r H = \Sigma \Delta_c H(\text{reactants}) - \Sigma \Delta_c H(\text{products})$ — the other way round

If your answer has the wrong sign for an obviously exothermic reaction, check this first. Arrowheads on the cycle are often a mark in themselves.

Bond enthalpy: $\Delta_r H = \Sigma(\text{bonds broken}) - \Sigma(\text{bonds formed})$. Draw the full displayed formula first — miscounting bond types is the usual error.

Why bond-enthalpy values are less reliable — both reasons: (1) mean bond enthalpies are **averaged over many different compounds**, not this molecule; (2) they apply to the **gaseous state only**, so state changes are ignored.

CALORIMETRY — $Q = mc\Delta T$, THE MOST REPEATED CALCULATION

1. $Q = mc\Delta T$, with **m = mass of the liquid being heated** (g), $c = 4.18 \text{ J g}^{-1} \text{ }^\circ\text{C}^{-1}$ · 2. convert to kJ and find moles of the limiting substance · 3. $\Delta H = -Q \div n$ · 4. sign, units (kJ mol⁻¹) and appropriate s.f.

Errors (other than heat loss, which is often excluded): incomplete combustion (soot/CO) · evaporation of the fuel before burning · heat used to warm the apparatus itself. Read what the question has already excluded — repeating it scores zero.

Reaction profiles: E_a is measured **from the reactants line to the peak** — never from zero, never from the products. A catalyst gives a **lower peak via an alternative pathway**; both curves must **start and end at identical levels**.

SHEET 2 — KINETICS & EQUILIBRIUM

MAXWELL-BOLTZMANN — THE FIVE FEATURES CHECKED EVERY TIME

1. Starts at the **origin** (no particle has zero energy) · 2. rises to a single peak · 3. long right tail that **approaches but never touches** the x-axis · 4. the **area is constant** (= total particles) at any temperature · 5. higher T → peak **lower, broader and shifted right** — never simply "up".

WHY TEMPERATURE RAISES THE RATE — ALL THREE LINKS, EACH A MARK

1. Raising T increases the **average kinetic energy** of the particles · 2. so a **greater proportion of particles have energy $\geq E_a$** · 3. so there is a greater frequency of **successful** collisions.

"More collisions happen" scores nothing — you must say **successful** collisions. And a catalyst does **not** change the curve: it moves the **E_a line left** on the same, unchanged curve.

Catalyst: provides an **alternative reaction pathway with a lower activation energy**, not used up. **Sustainability reasons:** allows a **lower operating temperature** for an acceptable rate → less energy and CO₂; a **selective** catalyst reduces by-products → better atom economy; it is **reused**, not consumed.

Rate from a graph: draw a **tangent** at the stated point, gradient = $\Delta y/\Delta x$, give units (cm³ s⁻¹ or mol dm⁻³ s⁻¹). **Show the construction lines on the graph** — the scheme requires it.

Change applied	Effect on the position of equilibrium
↑ concentration of a reactant	Shifts right
↑ pressure (gas equilibria)	Shifts to the side with fewer moles of gas
↑ temperature	Shifts in the endothermic direction
Catalyst	No effect on position or yield — equilibrium reached faster

ALWAYS SEPARATE RATE FROM YIELD

These are marked as **two different ideas**. Raising T for an exothermic reaction **increases the rate but decreases the yield**. "Faster" never automatically means "more product".

Industrial compromise: 1. low T raises yield (exothermic) but lowers rate → **moderate T compromise** · 2. high P raises both rate and yield (fewer gas moles on the right) **but is expensive** — thicker equipment, higher running costs → **compromise P** · 3. a catalyst raises the rate **without raising T**, protecting yield · 4. unreacted gases are **recycled**, so a low yield per pass is acceptable.

SHEET 3 — REDOX, GROUPS 1 & 2

Oxidation number rules, in priority order: uncombined element 0 · simple ion = its charge · sum = 0 (compound) or the charge (ion) · **Group 1 +1, Group 2 +2 · F always -1 · H +1 except -1 in metal hydrides · O -2 except -1 in peroxides and +2 in OF₂.**
Section A trap: oxygen in **H₂O₂ (-1)** and in **OF₂ (+2, fluorine outranks oxygen)**.

DISPROPORTIONATION — QUOTE THE NUMBERS OR LOSE MARKS

The **same element, in the same species**, is simultaneously oxidised **and** reduced.

Cold dilute NaOH: **Cl₂ + 2NaOH → NaCl + NaOCl + H₂O** — Cl goes **0 → -1** and **0 → +1**.

Hot conc. NaOH: **3Cl₂ + 6NaOH → 5NaCl + NaClO₃ + 3H₂O** — Cl goes **0 → -1** and **0 → +5**.

With water: **Cl₂ + H₂O ⇌ HOCl + HCl**.

"Chlorine changes oxidation state" without the numbers scores less than full marks.

Half-equations: balance the changing atom → balance O with **H₂O** → balance H with **H⁺** → balance charge with **e⁻**. To combine, scale so the electrons cancel, add, cancel.

Ion	Flame colour	Ion	Flame colour
Li ⁺	Red (crimson)	Ca ²⁺	Brick-red / orange-red
Na ⁺	Yellow / orange	Sr ²⁺	Red (crimson)
K ⁺	Lilac	Ba ²⁺	Green (apple / pale green)
Rb ⁺	Red	Cs ⁺	Blue / violet

HOW A FLAME COLOUR FORMS — ALL THREE STEPS ARE SEPARATE MARKS

Heat energy **excites/promotes electrons to a higher energy level** → the electrons **fall back down** to a lower level → **releasing energy as light of a specific wavelength/frequency**, giving the colour. "Electrons emit a colour" alone = 1 of 3.

THERMAL STABILITY (17/18 PAPERS) — AND THE DIRECTION TRAP

Down the group: **1.** the **cation gets larger** · **2.** so it has a **lower charge density and is less polarising** · **3.** so it **distorts the carbonate/nitrate anion less** · **4.** so the **covalent C-O (or N-O) bond inside the anion is weakened less** → a higher temperature is needed → **more thermally stable**.

Explicitly penalised: "the anion polarises the cation" (wrong way round) and "the **ionic** bond weakens" (it is the **covalent bond inside the anion**).

Decompositions: Group 2 carbonate → MgO + CO₂ · Group 2 nitrate → **2Mg(NO₃)₂ → 2MgO + 4NO₂ + O₂** (brown gas) · Group 1 nitrate (Na, K, Rb, Cs) → **2NaNO₃ → 2NaNO₂ + O₂** (no brown gas) · **LiNO₃ behaves like Group 2: 4LiNO₃ → 2Li₂O + 4NO₂ + O₂**, because Li⁺ is unusually small with a high charge density.

Solubility: Group 2 **hydroxides more soluble down; sulfates less soluble down** (BaSO₄ essentially insoluble — the basis of the sulfate test).

SHEET 4 — GROUP 7

Trend down the group	Reason
Electronegativity decreases	Larger atomic radius, so weaker nuclear attraction for a bonding pair despite higher nuclear charge
Boiling temperature increases	More electrons per molecule → stronger London forces → more energy to overcome
Oxidising power decreases	Larger atoms attract an incoming electron less strongly (further out, more shielded)

Halogen	In aqueous solution	In hexane / cyclohexane
Cl ₂	Pale green / colourless	Pale green / colourless
Br ₂	Orange	Orange
I ₂	Brown	Purple / violet

Iodine is the only halogen that changes colour between the layers — a very common Section A distractor gives it the same colour in both.

HALIDES WITH CONCENTRATED H₂SO₄ — LEARN THE WHOLE TABLE

All four first undergo a simple **acid-base (proton transfer) reaction** giving steamy/misty **HX fumes** — **this step is not redox**.

Cl⁻: misty fumes **only** — chloride cannot reduce H₂SO₄.

Br⁻: then **2HBr + H₂SO₄ → Br₂ + SO₂ + 2H₂O** — orange/brown vapour and choking SO₂. **S: +6 → +4.**

I⁻: then **8HI + H₂SO₄ → 4I₂ + H₂S + 4H₂O** (and 6HI + H₂SO₄ → 3I₂ + S + 4H₂O) — purple vapour / black solid, **yellow solid (S), bad-eggs smell (H₂S)**. **S: +6 → 0 or -2.**

The reasoning link: iodide is a stronger reducing agent because **I⁻ is larger, so its outer electron is lost more easily** — which is why it reduces sulfur furthest.

Silver nitrate test (acidify with dilute **HNO₃** first): **Cl⁻ white**, dissolves in **dilute NH₃** · **Br⁻ cream**, dissolves only in **concentrated NH₃** · **I⁻ yellow**, insoluble even in concentrated NH₃. **Ag⁺(aq) + X⁻(aq) → AgX(s).**

Bleach + strong acid is dangerous: the acid shifts **HClO ⇌ H⁺ + ClO⁻** and **HClO + Cl⁻ + H⁺ ⇌ Cl₂ + H₂O** to the right, releasing toxic chlorine. **You must reference the actual equilibria**, not just say "it's dangerous".

SHEET 5 — INTERMOLECULAR FORCES (every paper)

Force	Present between	Origin
London (dispersion)	All molecules and atoms, polar or not	A temporary dipole from momentarily uneven electron distribution induces a dipole in a neighbour
Permanent dipole-dipole	Polar molecules only	Permanent δ^+ and δ^- on neighbouring molecules attract
Hydrogen bonding	Only where H is bonded directly to N, O or F	Very strong permanent dipole between the δ^+ H and a lone pair on N/O/F

THREE HYDROGEN-BONDING CRITERIA — ALL MUST BE TRUE

1. H bonded **directly** to N, O or F · **2.** that atom has a **lone pair** · **3.** the bond is highly polar because N/O/F is **small and very electronegative**.
Can: water, alcohols, ammonia, amines, carboxylic acids (anything with $-\text{OH}$, $-\text{NH}$, $-\text{NH}_2$). **Cannot:** ketones, aldehydes, ethers, esters — **they contain oxygen but no H bonded to it**.
The HF trap: HF hydrogen bonds but boils far below water, because **each water molecule can form twice as many hydrogen bonds** (2 lone pairs + 2 O-H, versus HF's effective 1). Compare the **number per molecule**, not just whether H-bonding is present.

THE BOILING-POINT ANSWER STRUCTURE — USE IT EVERY TIME

1. Identify which force(s) each substance has (**all** have London) · **2. London forces increase with more electrons** — use this within a subgroup · **3. hydrogen bonding is much the strongest**, so an H-bonding substance has an anomalously high b.p. even with fewer electrons · **4.** conclude: **stronger forces → more energy to overcome → higher boiling temperature**.

Solubility — like dissolves like: dissolving works when the **solute-solvent forces formed** are of similar strength to the **solute-solute and solvent-solvent forces broken**. Short alcohols dissolve in water (hydrogen bond to it); **long-chain alcohols less so**, as the non-polar chain becomes a larger fraction; non-polar iodine dissolves in hexane, not water.

SHEET 6 — HALOGENOALKANES (18/18 papers)

	S_N2 — primary	S_N1 — tertiary
Steps	One step , no charged intermediate	Two steps via a carbocation
Arrows	Lone pair on OH ⁻ → the δ^+ carbon (attack from the side opposite the leaving group); C-X bond → the halogen	Step 1: C-X bond → halogen, giving a planar carbocation + X ⁻ . Step 2: lone pair on OH ⁻ → the C⁺
Why	Little steric hindrance	Stable tertiary carbocation; the carbocation is trigonal planar

Mechanism errors that lose marks: missing **lone pair** on the nucleophile · a **single-headed "fish-hook"** arrow instead of a full curly arrow · an arrow starting **from the atom itself** rather than from a bond or lone pair · missing the **δ^+/δ^-** on C-X.

Secondary halogenoalkanes react by **either** mechanism, which is why questions say the rate "cannot be predicted" for them.

HYDROLYSIS RATE — THE MOST REPEATED "DO NOT AWARD" TRAP

Rate order is always **iodo > bromo > chloro**, and the explanation is **entirely bond enthalpy**: down Group 7 the **C-X bond gets longer and therefore weaker**, so it **breaks more easily**.

Do NOT explain it with electronegativity or bond polarity — C-Cl is the **most** polar yet reacts **slowest**, so polarity cannot be the deciding factor.

Separately, **tertiary hydrolyses faster than primary** with the same halogen — that is because S_N1 has a lower E_a, **not** bond enthalpy (it is the same C-Br in both).

THE SOLVENT DECIDES — SUBSTITUTION vs ELIMINATION

Aqueous KOH/NaOH, reflux → **substitution**; OH⁻ acts as a **nucleophile**; product is an **alcohol**.

Ethanolic KOH, reflux → **elimination**; OH⁻ acts as a **base**; product is an **alkene**.

Elimination arrows: OH⁻ lone pair → the **β -hydrogen** · the **C-H bond** → between the two carbons (forming C=C) · the **C-X bond** → the halogen.

If more than one β -carbon has a removable H, **more than one alkene** forms — including E/Z isomers.

Other reactions: KCN in ethanol, reflux → **nitrile**, which **adds one carbon to the chain** (the accepted reason it matters in synthesis) · **excess concentrated ethanolic NH₃, heated in a sealed tube** → **primary amine** (excess minimises further substitution).

SHEET 7 — ALCOHOLS & SPECTROSCOPY

Alcohol	Product and apparatus	Why
Primary	Distil off as it forms → aldehyde. Reflux with excess → carboxylic acid	Distillation removes the aldehyde before further oxidation; reflux keeps it in contact with the oxidant
Secondary	Reflux → ketone	That carbon has no second H to lose
Tertiary	No reaction — stays orange	No H on the carbon bearing -OH

Oxidant: $\text{K}_2\text{Cr}_2\text{O}_7$ acidified with dilute H_2SO_4 ; colour change orange → green.

"Do not award reflux" appears for the aldehyde step, and "do not award distillation" for the acid step — mixing them up, or just writing "heat", loses the mark.

Distinguishing the products: Fehling's/Benedict's → aldehyde gives a brick-red precipitate, ketone stays blue · or IR: the broad alcohol O-H diminishes and a sharp C=O appears.

Alcohol → halogenoalkane: PCl_5 at room temperature (also the test for -OH: misty/steamy white fumes of HCl) · NaBr/KBr + conc. H_2SO_4 (HBr in situ) · red phosphorus + I_2 .

Dehydration: concentrated H_3PO_4 or H_2SO_4 , heat → alkene. If the alcohol can lose H from more than one side, more than one alkene forms.

Bond	Wavenumber / cm^{-1}
O-H (alcohol)	3750-3200, broad
O-H (carboxylic acid)	3300-2500, very broad, overlapping C-H
N-H	3500-3300
$\text{C}\equiv\text{N}$	2260-2220
C=O	1740-1700
C=C	1669-1645 (often weak)

THE "DISTINGUISH TWO COMPOUNDS BY IR" TEMPLATE

State which peak is **present in one and absent in the other**, with **both the wavenumber range AND the bond, for each compound**: "the alcohol shows a broad O-H at 3750-3200 cm^{-1} which is absent in the aldehyde; the aldehyde shows a C=O at 1740-1700 cm^{-1} which is absent in the alcohol."

"The spectra are different" scores nothing. **Never base an answer on the fingerprint region** below ~1500 cm^{-1} , and ignore C-H — it is in virtually everything.

Mass spectrometry: M^+ is the peak at the highest m/z and equals M_r . Fragments: 15 CH_3^+ · 29 CHO^+ or C_2H_5^+ · 43 CH_3CO^+ or C_3H_7^+ · 45 COOH^+ or CH_3CHOH^+ .

SHEET 8 — PRACTICAL & QUANTITATIVE SKILLS

Quantity	Formula
Moles from mass	$n = \text{mass} \div \text{M}_r$
Moles from concentration	$n = c \times V$, with V in dm^3 ($\text{cm}^3 \div 1000$)
Moles of gas at r.t.p.	$n = V(\text{cm}^3) \div 24\,000$, or $V(\text{dm}^3) \div 24.0$
Moles of gas, any conditions	$pV = nRT$ — p in Pa, V in m^3 , T in K, R = 8.31
Percentage yield	$(\text{actual} \div \text{theoretical}) \times 100$
Atom economy	$(\text{M}_r \text{ of desired product} \div \Sigma \text{M}_r \text{ of all products}) \times 100$

$pV = nRT$ — THREE CONVERSIONS, EACH A COMMON LOST MARK

kPa → Pa ($\times 1000$) · $\text{cm}^3 \rightarrow \text{m}^3$ ($\div 10^6$) · $^\circ\text{C} \rightarrow \text{K}$ (+273)

Forgetting any one loses the mark even when the rest of the method is right.

Titration: moles of the known → mole ratio from the equation → scale up ($\times 10$ if a 25.0 cm^3 sample came from a 250 cm^3 flask) → answer.

Concordant titres agree within 0.10 cm^3 ; average **only** concordant ones, discarding the rough. A burette reading is $\pm 0.05 \text{ cm}^3$, so a **titre is ± 0.10** (two readings).

Rinsing: burette and pipette with **the solution they will contain**; conical flask with **distilled water only**.

Empirical formula: mass or % $\div A_r \rightarrow \div \text{smallest} \rightarrow \text{whole-number ratio} \rightarrow \text{molecular formula by comparing with } \text{M}_r \text{ (from } \text{M}^+ \text{)}.$

SHEET 9 — THE SIX QER ARCHETYPES

The 6-mark question is drawn from a small repeating set. The substances change; the chemistry points do not.

Archetype	Content template
A — Boiling-temperature pattern	1. Name the force(s) in each (all have London) · 2. within a London-only subgroup, more electrons → stronger · 3. hydrogen bonding is much the strongest · 4. identify which can H-bond (H on N/O/F) and conclude they are anomalously high · 5. tie-breaker between two H-bonders = number of hydrogen bonds per molecule
B — Industrial conditions	1. Low T raises yield (exothermic, Le Chatelier) · 2. but lowers rate → compromise T · 3. high P raises rate and yield (fewer gas moles) · 4. but is expensive → compromise P · 5. catalyst raises rate without raising T; unreacted gases recycled
C — Substitution vs elimination	1. Both use OH ⁻ and heat · 2. aqueous → substitution, OH ⁻ as nucleophile , alcohol — give the equation · 3. ethanolic → elimination, OH ⁻ as base , alkene — give the equation · 4. name/draw any alkene isomers, noting E/Z
D — Conc. H₂SO₄ with two halides	1. Both first give an acid-base, non-redox reaction with misty HX fumes — equation · 2. the further redox products for each (Br ₂ /SO ₂ ; I ₂ /S/H ₂ S); chloride shows none · 3. the oxidation number change of sulfur in each case · 4. link to structure: the larger halide loses its outer electron more easily , so is the better reducing agent
E — Lithium / Group 1 vs Group 2	1. Li⁺ and Mg²⁺ have similar ionic radii · 2. equations for both decompositions and for NaNO ₃ (nitrite + O ₂ only) for contrast · 3. the small, high-charge-density ion polarises the nitrate more , weakening the N-O bond enough for full decomposition
F — Devise an experiment to find an A_r	1. Weigh the container, then with the solid · 2. heat to constant mass to ensure decomposition is complete · 3. moles of gas lost from the mass loss · 4. M _r of the compound, then subtract the known non-metal part to get A _r and identify it

IF THE QUESTION GIVES BULLET-POINTED SUB-REQUIREMENTS

Those bullets are **almost always a direct map of the indicative content**. Answer them **in the order given, one clear point per bullet**, and you will hit close to the maximum content mark.

Include at least one equation whenever specific reactions are mentioned — a correctly balanced equation is nearly always one of the points on offer.

SHEET 10 — THE REJECTIONS BANK

Do not write	Write instead
A definition without " 1 mole " or without standard states	Both, every time
An equation with no state symbols when asked for	(s) (l) (g) (aq) — a whole mark
An enthalpy answer with no sign or no units	Sign and kJ mol ⁻¹ — separately awarded
"Heat loss" when the question has excluded it	Incomplete combustion · fuel evaporation · heat warming the apparatus
"More collisions happen"	A greater proportion of particles have E ≥ E_a , so more successful collisions
A catalyst "increases the energy of the particles"	It provides an alternative pathway of lower E_a ; the curve is unchanged, the E_a line moves left
Treating rate and yield as the same thing	They are marked as two separate ideas
"Chlorine changes oxidation state"	The actual numbers: 0 → -1 and 0 → +1
"The anion polarises the cation"	The cation polarises the anion
"The ionic bond weakens"	The covalent bond inside the carbonate/nitrate ion weakens
Explaining hydrolysis rate by electronegativity or polarity	Bond enthalpy only — C-I is longest and weakest
"KOH" without saying aqueous or ethanolic	Aqueous → substitution; ethanolic → elimination
"Heat" for an alcohol oxidation	Distil off as it forms (aldehyde) or reflux (acid) — never the reverse
A bond name with no wavenumber range , or a range with no bond	Both, from the Data Booklet
An answer based on the fingerprint region	Only above ~1500 cm ⁻¹ unless the question allows otherwise
A hydrogen-bonding claim for a ketone, aldehyde, ether or ester	They contain O but no H bonded to it — no H-bonding between their own molecules
A QER answer as a disconnected list	5–6 points linked with <i>because / therefore / this means</i>
An extra statement you are unsure of in a QER	Leave it out — wrong chemistry costs the reasoning marks

WALKING INTO THE EXAM

Section A in about 22 minutes — it is a fixed 20 marks and the cheapest on the paper. Check **state symbols, sign, units and significant figures** on every calculation line. Never leave a number blank, because **TE runs through almost every multi-step calculation**. On the 6-mark question, bullet five or six points first, **then** add the connectives.